

Definition: Product is expected to deliver the shortest path to the nearest bicycle parking lot to a desired location, ensuring that the user has a bicycle parking lot to park at, given the weather, time and capacity.

Product offers routes to parking lots and filters such as sheltered parking lots, group riding and sharing of ride information for safety. Report system is also offered for users to give feedback about the capacity and availability of the parking lot.

Product focuses on user's convenience, allowing them to save time and not manually look for a parking lot. It also saves time by making sure it's the parking lot they need, for example, sheltered or not sheltered.

Data Dictionary (need to wait for requirements to be out)

- Define the terms used (users, and objects)
- Name (String) (size 20) Name of User, Eg: Linyue
- Address (String) (Size 50) Addy, Eg: 126 Sembawang Road
- Phone No (Text) (size 10)
- DOB
- Locations (altitude, latitude)

Name: User

- Type: Class
- Description: An entity that represents a human actor that interacts with this system. This entity begins by looking for a bicycle parking spot within the geographic region of Singapore.
- Source: User database
- Format: Object type consisting of phone number(s), firstName, lastName, age, gender as properties.
- Relationships: Contains saved locations (coordinates) and/or addresses that the User may use during their interactions within the system.
- Ownership: Maintained by the User. Relevant properties may only be updated by the User that owns this data.
- Access Control: Only the original User may access and/or modify the object data, potentially through SMS verification.

Name: Cyclist

- Type: Object
- Description: An instance of the User class that represents a human actor with the role of a 'cyclist', defined as someone who actively cycles or performs cycling activities within the geographic region.
- Source: User class

- Format: User class with further properties including (but not limited to) the endurance and range of the actor.
- Relationships: Behaves as an entity of User so therefore, contains relationships in the same way.
- Ownership: Similar to the User class.
- Access Control: Similar to the User class.

Name: Maps/Geographic Region, or Coordinates

- Type: API reference, Object/String
- Description: Live referencing of mapping systems using relevant web APIs.
- Source: Owners of the APIs.
- Format: JSON/XML return types with usable properties/notations.
- Relationships: Interacts with the live dataset to customize markers/information on mapping services.
- Ownership: System-referenced through a unique API key, potentially.
- Access: Fetch API with API key verification to access the services.

Name: Land Transport Authority (LTA)

- Type: Entity
- Description: The governing authority of Singapore's land transport systems.
- Source: LTA's DataMall
- Format: See above.

Functional Requirements:

Locate Nearest Bicycle Parking Lot:

- Users should be able to initiate a search for the nearest bicycle parking lot to the user by providing their current coordinates.
- The system will utilize these coordinates to identify the closest parking facility to them based

Find Bicycle Parking Lot Near Specified Location:

- Users can input the coordinates of a desired location they intend to visit.
- Upon submission, the system will determine the nearest bicycle parking lot to the specified destination.

Display Default Top 3 Results:

- Upon search initiation, the system will automatically present the top 3 nearest bicycle parking lots.
- The system map display will display the location of these top 3 nearest bicycle lots.

Option to See More Results:

- Users can opt to view additional bicycle parking options beyond the initial 3 by selecting a designated "See More" button.

Detailed Information Display:

- Upon selecting a specific parking lot, users will be presented with comprehensive details such as slot availability and shelter availability of the specific parking lot.

Filtering Options:

- Users can filter bicycle parking lots based on attributes such as availability, and shelter availability.
- The system will then display only the parking options that meet the specified criteria to the user.

Select Destination:

- After reviewing available options, users can select a preferred bicycle parking lot as their destination.

Map Display:

- A dynamic map display will showcase the user's current location, the destination bicycle parking lot, and other bicycle parking options.

Integration with Google Maps:

- The system will fetch live data from Google map and then display the map accordingly on the map display.
- The system will use Google Maps GPS information to design the navigation directions for the user.

Navigation Directions:

- Upon selecting a bicycle parking lot as the destination, users will receive navigation directions from their current location to destination

Save Common Inputs:

- Users can save frequently visited locations such as home and workplace for quick access in future searches.
- The system will have a database to save these user inputs for each user.

Parking Lot Occupancy Reporting:

- Users have the option to report the occupancy status of a bicycle parking lot, indicating whether it is fully occupied or not.

Real-Time Updates:

- Users can contribute real-time updates on parking lot occupancy, helping to maintain accurate and up-to-date information within the system.

Integration with LTA API:

- The system will integrate with the Land Transport Authority (LTA) API to fetch live data regarding bicycle parking lots.

Non-functional Requirements:

Performance:

- Response time for fetching and displaying parking lot information should be within 3 seconds.

Usability:

- The user interface should be intuitive and easy to navigate for more seamless user experience.
- Information presented to users should be clear, concise, and easily understandable in their language of choice.
- The system should support accessibility features to accommodate users with reading disabilities such as text to speech.

Reliability:

- The system should operate reliably under normal usage conditions, with minimal downtime or disruptions at a maximum of 0.5% downtime per day.
- Data presented to users should be accurate and up-to-date based on the latest update on LTA API and google maps, reflecting real-time information about parking lot availability.

Security:

- User data, including location information and saved inputs, should be securely stored and protected against unauthorized access or manipulation.
- Communication between the user's device and the system should be encrypted to ensure data privacy.

Scalability:

- The system should be able to handle a growing user base of at least Singapore's current population and increasing data volume without significant degradation in performance.

Maintainability:

- The system should be modular and well-organized, facilitating ease of maintenance and future enhancements.
- Code should be well-documented and adhere to coding standards to support ongoing development efforts.

Compatibility:

- The system should be compatible with a wide range of mobile devices and web browsers to ensure broad accessibility.
- Integration with external APIs, such as Google Maps and the LTA API, should follow industry standards and protocols.

Availability:

- The system should have high availability, with uptime of at least 99.5% to ensure continuous access for users.

Interoperability:

- The system should be interoperable with other transportation and navigation services to provide seamless user experiences.
- Integration with external APIs should be standardized to enable interoperability with third-party applications and services.

Performance Under Load:

- The system should be able to handle at least 10,000 concurrent user requests and heavy loads during peak usage periods without degradation in performance.
- Load testing should be conducted periodically to identify and address potential performance bottlenecks.

Data Privacy and Compliance:

- The system should comply with relevant data privacy regulations, such as PDPA to protect user privacy and rights.
- User consent should be obtained for collecting and processing personal data, including location information.

Use cases (need to wait for requirements to be out)

- Perform a search query.
- Split into single options (user filter for each criteria, extending from search)
- View a list of available spots.

UI:

Main page is a login page with phone number verification and pw

-> lead u to a map like gmaps with a search bar at the top and a filter bar below with options(radius, shelter, total lots, type of lot)

When user uses search bar, there's an option displayed below the bar to show saved location

-> when he clicks on it, it will show a list of saved locations and when he selects one, the location will be inputted into the search bar

-> if not he can just key into the search bar

Filter bar will display dropdown selections for radius, shelter, type of lots, total lots(can be displayed in range, eg. 1-10, 11-20, 21-30)

Once search is done, map will display location of all applicable nearest lots on map

-> directions button bottom left, and a display current location button right

-> on GPS prompt if not on

-> if on show current location represented by dot

Left panel (once clicked) displays share location, share group ride destination